AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 21 · STUDENT LESSON

Similarity by Selected Traits

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

Week 9: selected features determine what similarity means.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Represent items with numeric features. Try the recall before opening notes.

Your learning target

I can represent items with numeric features.

I can calculate a simple **distance**.

I can explain how units and chosen features affect **similarity**.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Repair Bias in Data. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Choose Informative Questions. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 21: **Similarity** by Selected Traits

Different feature choices produce different notions of closeness.



THINKER MISSION

Compute simple **distance** on genre, length, and difficulty.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

What does abs(3 - 7) measure? If unfamiliar, calculate the positive gap.

This week’s end goal

By the end, I can calculate a feature **distance** and state how scaling changes its meaning.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Similarity by Selected Traits case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use **similarity** by selected traits to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

A fictional activity chooser represents each activity with energy and group-work ratings on a 0 to 5 **scale**. A learner requests [2, 4]. We compare options by adding absolute differences.

This is a chosen rule for **similarity**; it is not a **learned model** or an objective measure of quality.

STUDENT LESSON · 3 OF 18

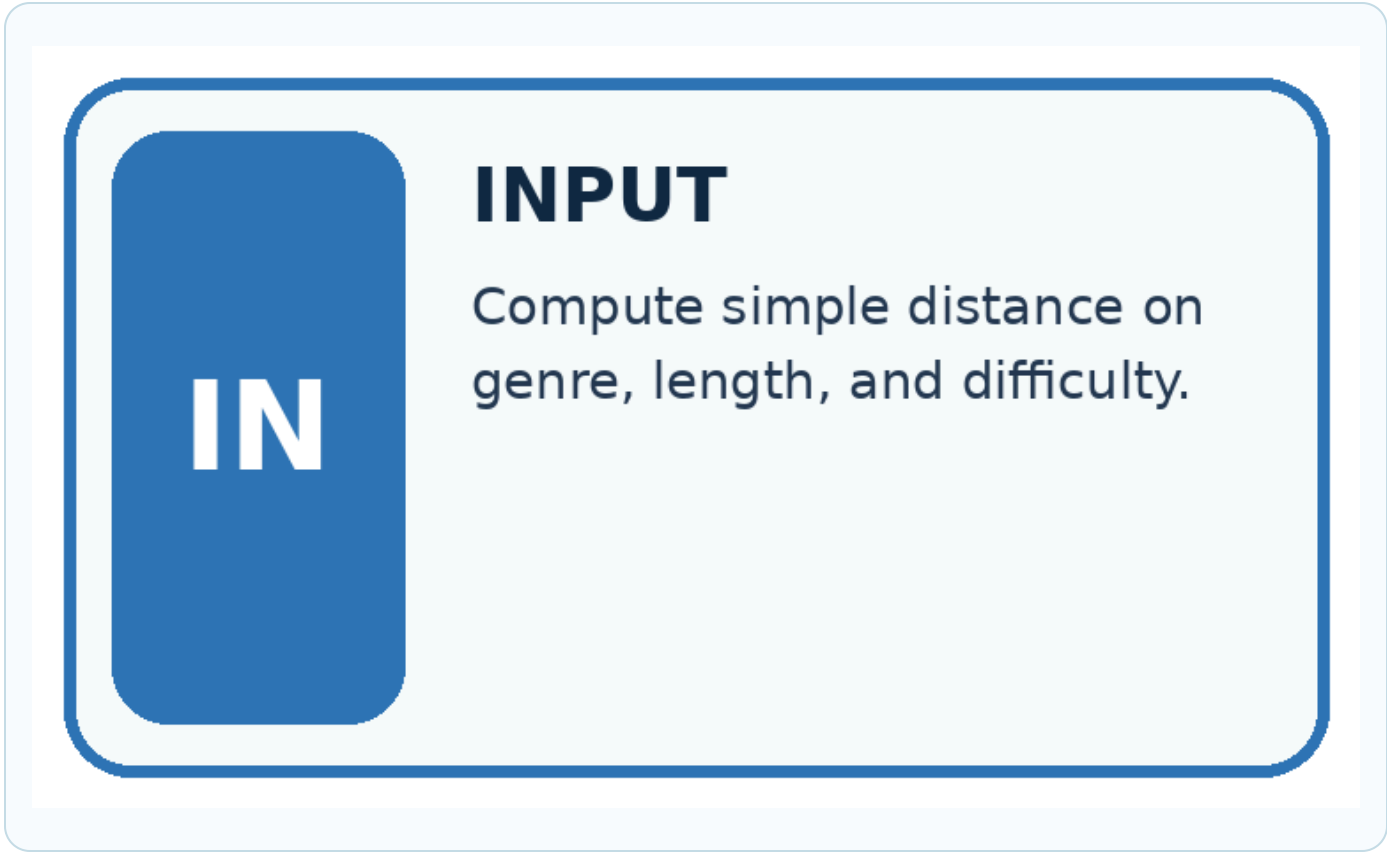
See how it works

> Follow the mechanism for Similarity by Selected Traits. Connect each labelled input, deciding step and result.

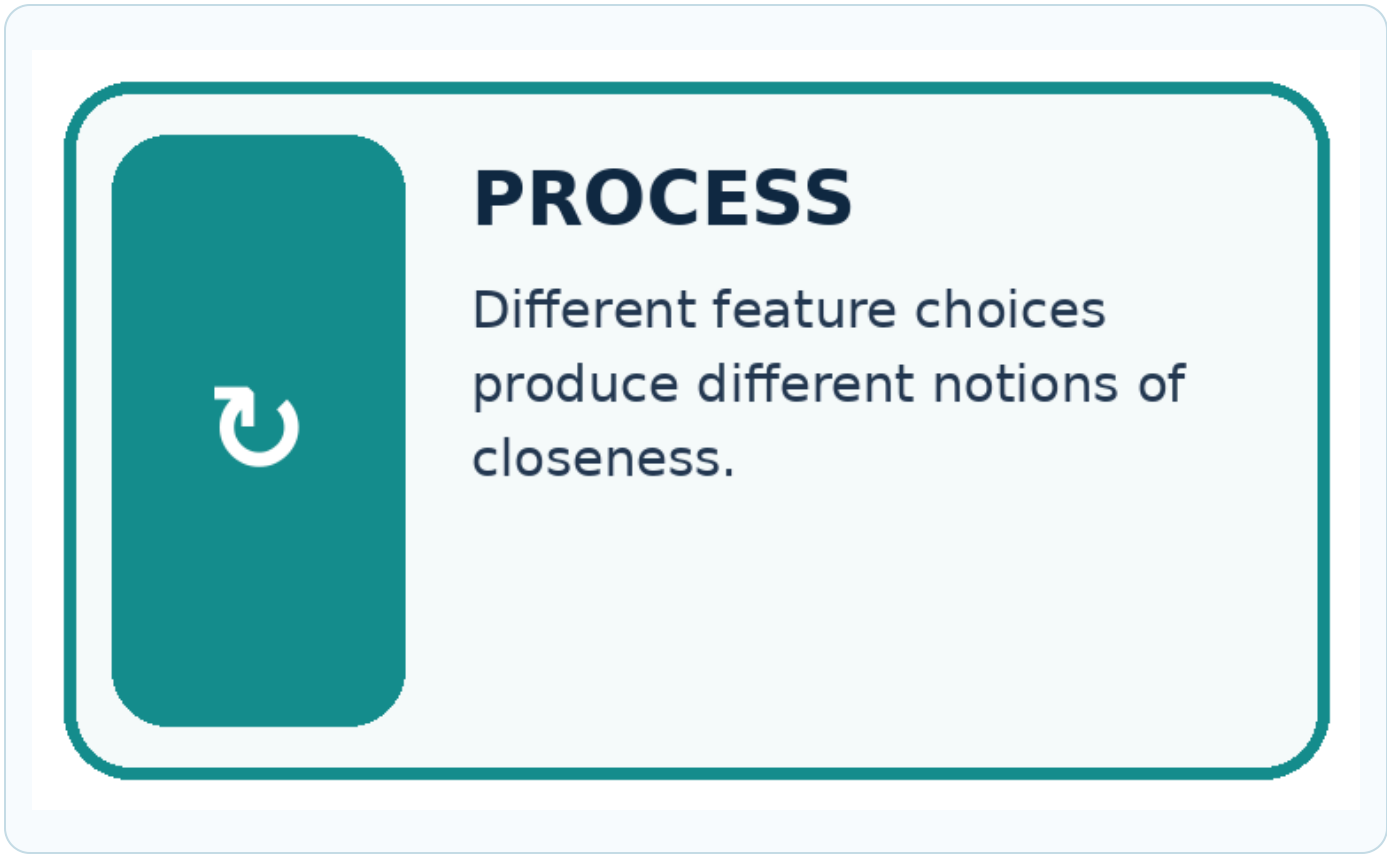
VISUAL MECHANISM

Input → **Process** → **Output**

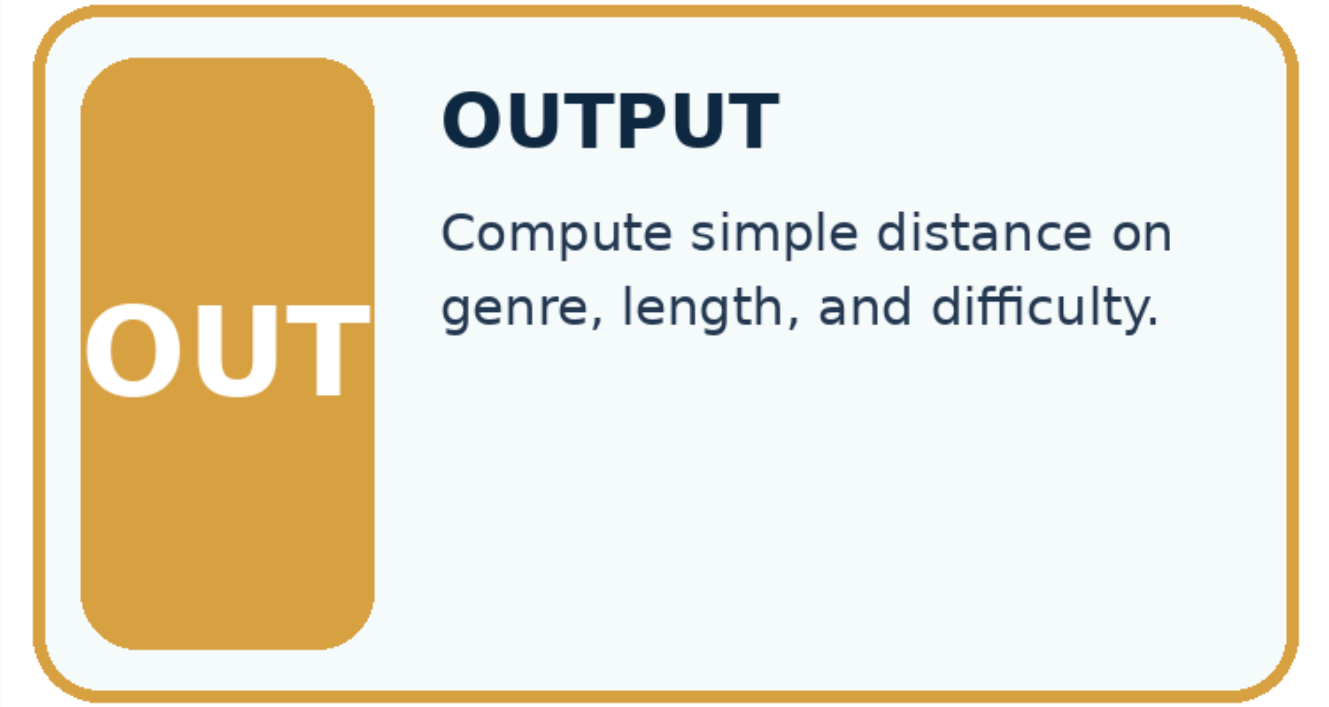
Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Compute simple distance on genre, length, and difficulty.

↓


PROCESS
Different feature choices produce different notions of closeness.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

```
target = [2, 4]
options = {"Sketch": [1, 3], "Relay": [5, 4]}
for name in options:
    features = options[name]
    distance = (abs(features[0] - target[0])
                + abs(features[1] - target[1]))
    print(name, distance)
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Sketch [1, 3]	1 + 1	Distance 2
Relay [5, 4]	3 + 0	Distance 3

Read each row from left to right and connect the evidence to the stated rule.

abs gives a nonnegative gap. Under this rule Sketch is closer. Feature order must match: first energy, then group work. Both **scale**s are 0 to 5, but that does not prove one unit has identical practical meaning in both features.

Calculate a gap one feature at a time

Keep the features in the same order: energy first, group work second. For target [2,4] and Sketch [1,3], the gaps are abs(1-2)=1 and abs(3-4)=1; the **distance** is 2. Relay [5,4] has gaps 3 and 0, so its **distance** is 3.

Smaller means closer under this particular rule. It does not mean better in every way, because the two recorded features leave out other needs.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

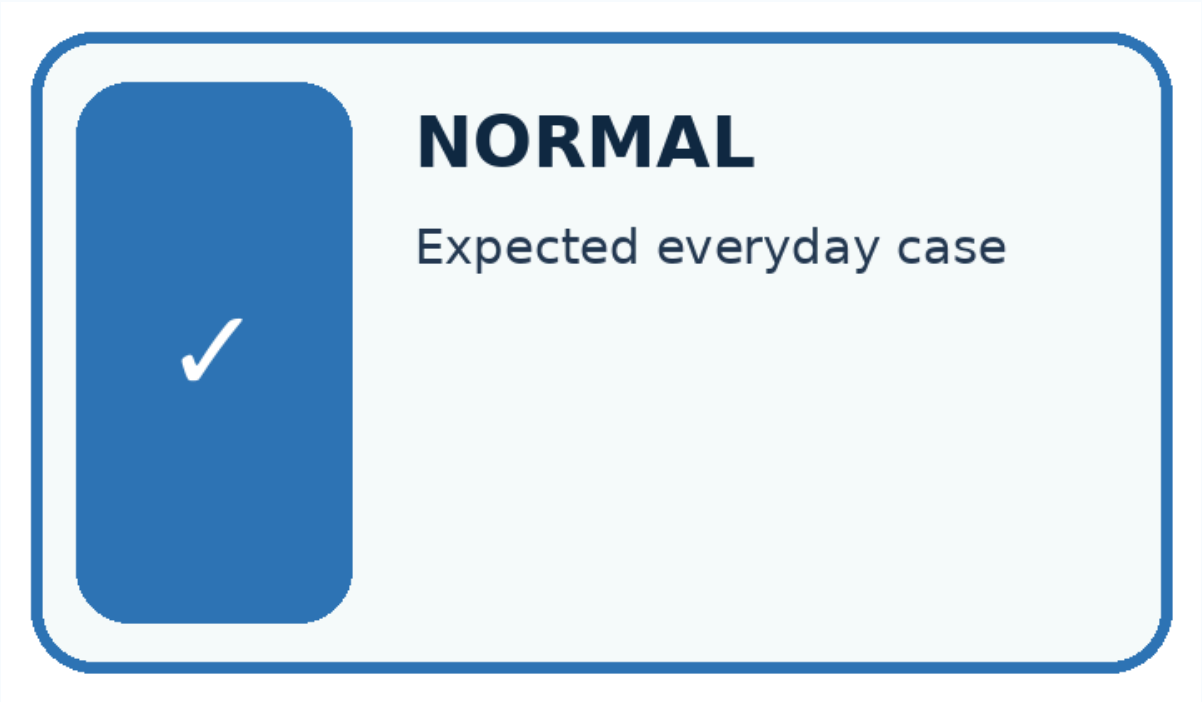
> Compare the supplied cases for Similarity by Selected Traits. Identify the detail that changes

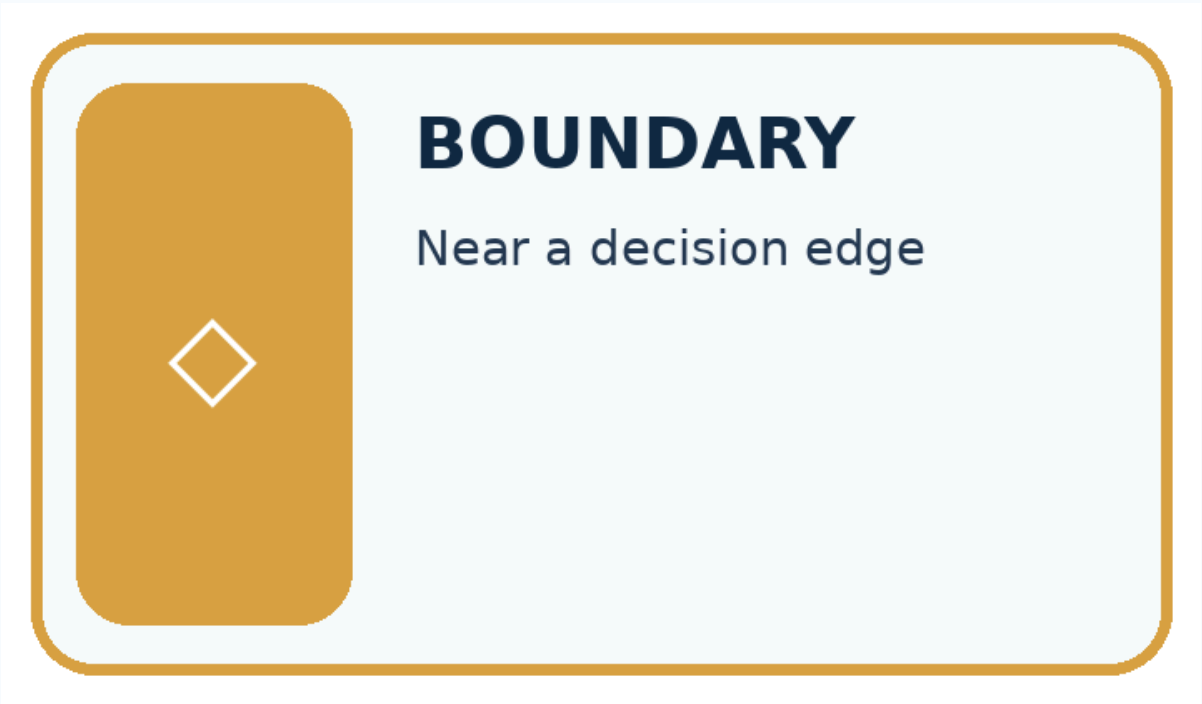
> the conclusion.

> **The case to investigate**
>
> CASE GALLERY

Four Tests, Four Kinds of **Evidence**

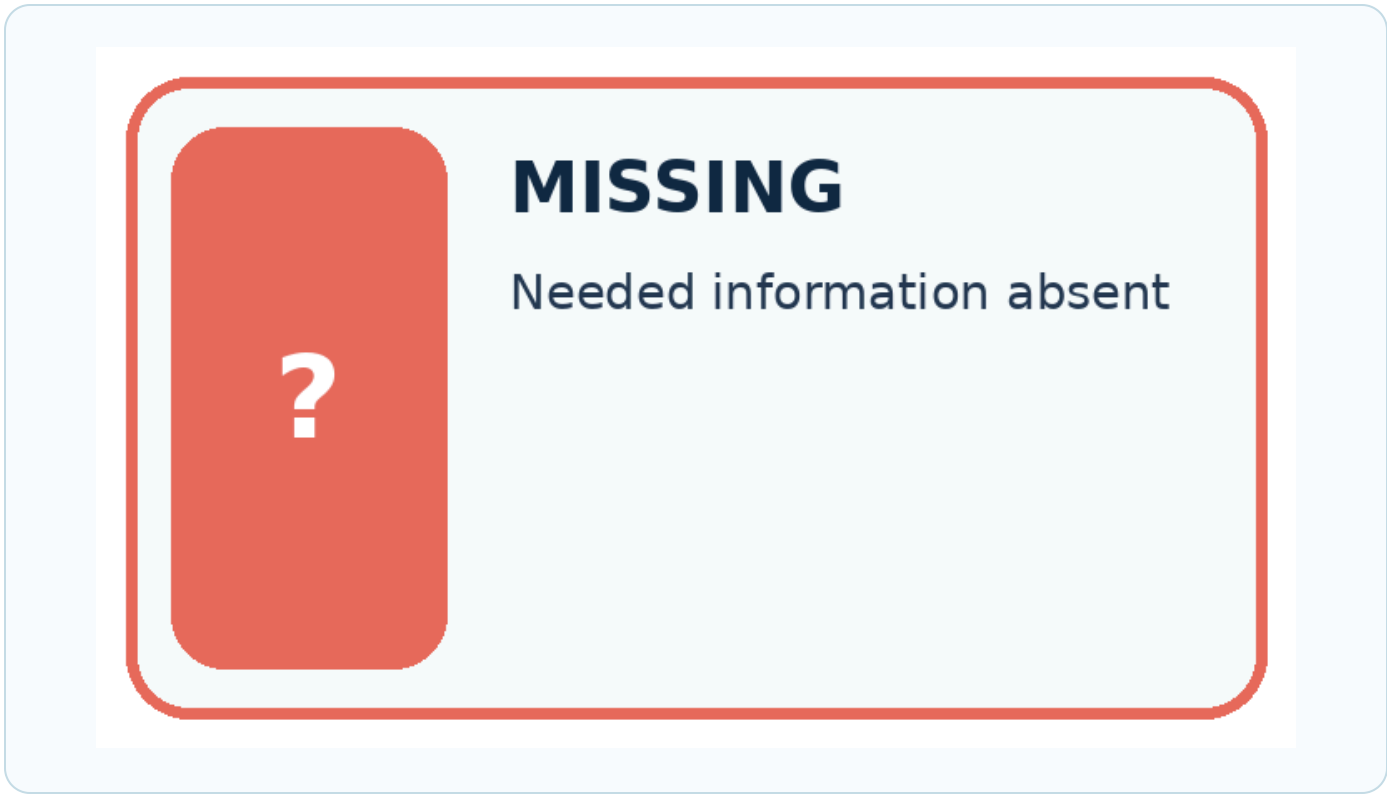
A system is not understood until it survives more than one comfortable example.





The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.


?
MISSING
Needed information absent


→
TRANSFER
Fresh situation or user

Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Give the first-**coordinate** difference weight 2 while the second keeps weight 1.

Predicted result

Prediction: **Distance**s become A=3, B=4, C=3; A and C tie.

Why the result follows

Reason: The representation's comparison rule changes; the supplied vectors stay the same.

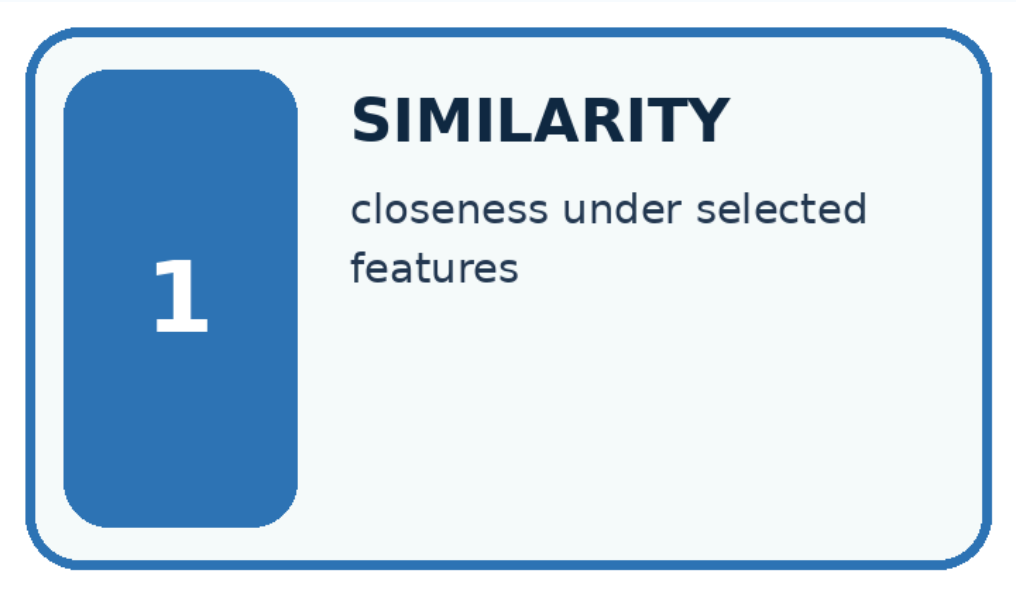
Words that unlock the idea

> Use the vocabulary of Similarity by Selected Traits to explain a concrete decision, rather than only memorising definitions.

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

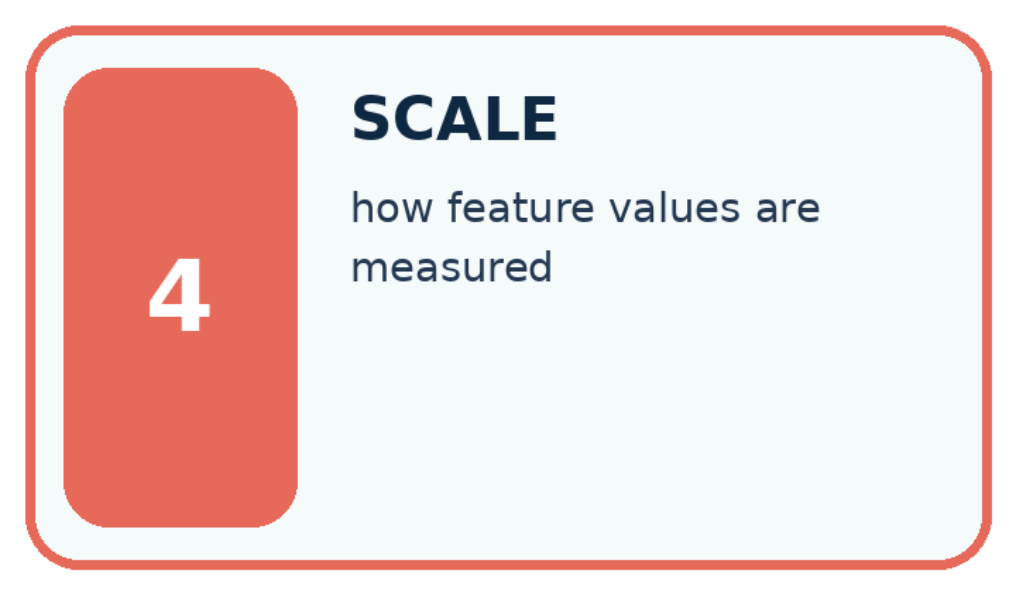
DISTANCE

number measuring difference

3

COORDINATE

feature value used as a position



Word	My own example	Not this / boundary
similarity	How alike two items are according to selected features and a declared comparison method.	
distance	A specified measure of how far apart two representations are.	
coordinate	A value identifying position along an agreed axis or within a space.	
scale	The units and range used to represent or compare a quantity.	

Words used in the closer explanation

Word	Meaning
Feature vector	An ordered list of numeric features
Manhattan distance	The sum of absolute feature differences
Normalization	Rescaling features to a comparable scale

STUDENT LESSON · 6 OF 18

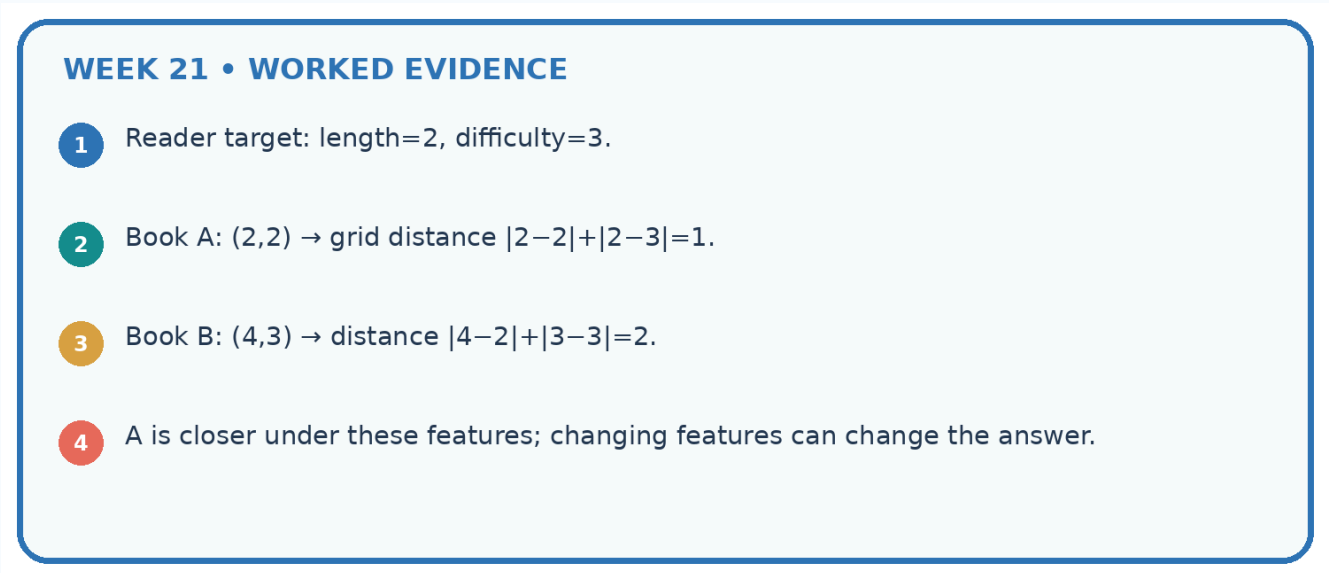
Follow a worked example

> Follow the worked trace for Similarity by Selected Traits, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Reader target: length=2, difficulty=3.

2. Book A: (2,2) → grid **distance** |2−2|+|2−3|=1.

3. Book B: (4,3) → **distance** $|4-2|+|3-3|=2$.

4. A is closer under these features; changing features can change the answer.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Find the distance from target [2, 4] to [2, 4] and to [3, 3].

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Similarity by Selected Traits

A fictional toy book matcher represents two preferences with numeric features.

The facts we will use

Supplied facts

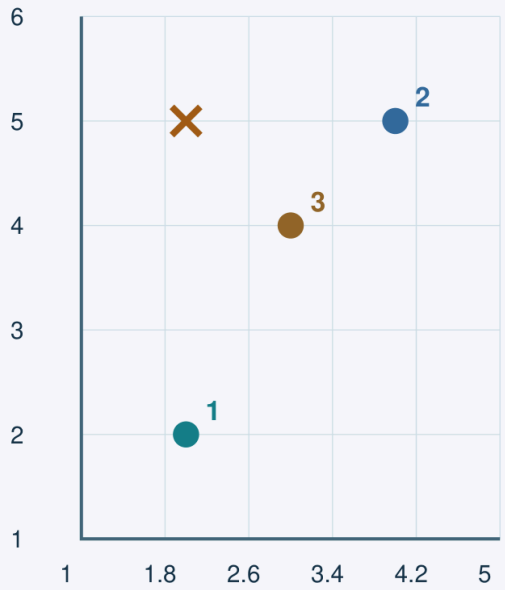
Target vector [2,5]. Candidate A=[2,2], B=[4,5], C=[3,4].

Supplied facts

Use **Manhattan distance**: add absolute **coordinate** differences; return all smallest-**distance** candidates.

AI ACADEMY / AI-2/21

Toy feature positions



Horizontal: first toy feature; vertical: second toy feature

1: A at (2, 2)

2: B at (4, 5)

3: C at (3, 4)

Cross: Target at (2, 5)

Distance rule: Manhattan distance; horizontal=first feature, vertical=second feature

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/21

Similarity by Selected Traits

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Distance A

$|2-2|+|5-2|=0+3=3$.

2. Distance B

$|2-4|+|5-5|=2+0=2$.

3. Distance C

$|2-3|+|5-4|=1+1=2$.

4. Choose closest

B and C tie at distance 2; keep the tie visible.

CHANGE ONE THING

Give the first-coordinate difference weight 2 while the second keeps weight 1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Distance A	$\lvert 2-2\rvert+\lvert 5-2\rvert=0+3=3$.
2. Distance B	$\lvert 2-4\rvert+\lvert 5-5\rvert=2+0=2$.
3. Distance C	$\lvert 2-3\rvert+\lvert 5-4\rvert=1+1=2$.
4. Choose closest	B and C tie at distance 2; keep the tie visible.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Each **coordinate** contributes according to the declared **distance** rule. Smaller **distance** means closer in this representation, not automatically a better reading experience.

If one **coordinate** uses a much larger numerical **scale**, it can dominate the sum even when it is not more important. Feature units and scaling belong in the method record.

What this example does not establish

Keep the stated limit

A toy numeric encoding does not establish meaningful interpersonal **distance**s or suitability of a recommendation.

Keep the stated limit

Read the labelled picture: Explain why B ceases to be a closest option without any candidate data changing.

Keep the stated limit

Change one thing: Give the first-**coordinate** difference weight 2 while the second keeps weight 1. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Similarity by Selected Traits claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Similarity is an objective fact independent of chosen features. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Similarity depends on the features collected and the **distance** rule. Missing accessibility, cost or preference information can make the closest item unsuitable. Hard requirements should be checked separately.

> **Two activities have distance zero. Must they suit every learner equally well?**

STUDENT LESSON · 8 OF 18

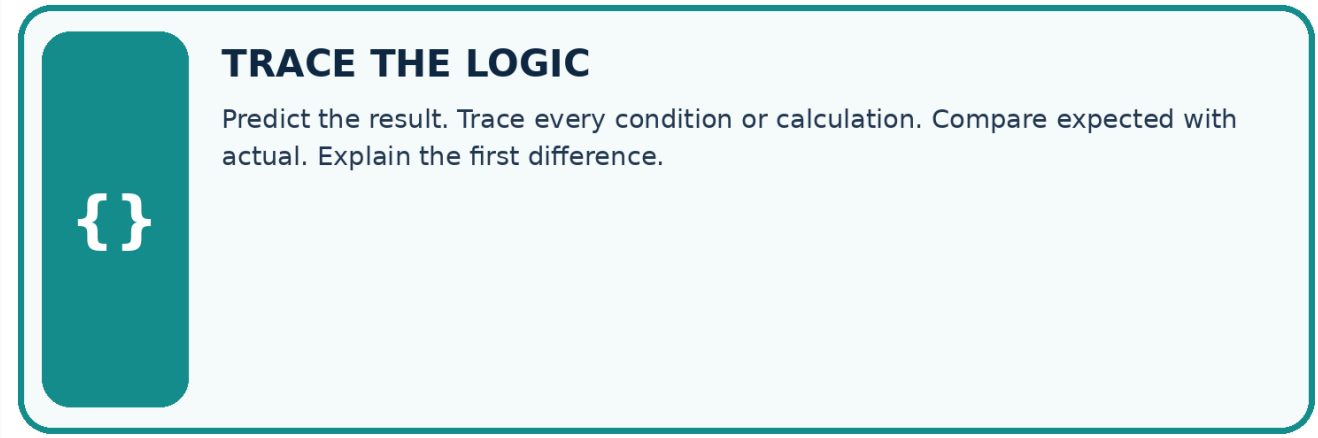
Practise together

> Practise Similarity by Selected Traits with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

> Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

If one feature is duration in minutes up to 60 and another a rating up to 5, raw **distance** may be dominated by duration. Specify scaling or weights before ranking; do not silently change units to get a preferred winner.

Practise with less help: Compute distances

Use workbook W2 for the connected example “**Distance** between **feature vector**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Distance** between **feature vector**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Explain why B ceases to be a closest option without any candidate data changing.

> **My explanation using a pictured step and the given facts**
>
> _____
>
> _____
>
> _____
>
> _____

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

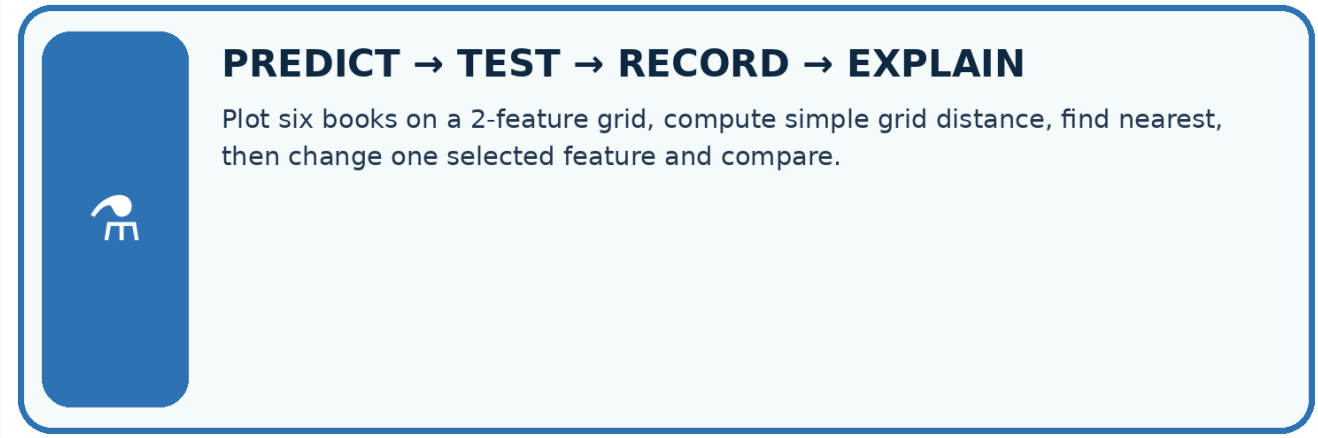
Run the investigation

> Investigate Similarity by Selected Traits. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Plot six books on a 2-feature grid, compute simple grid **distance**, find nearest, then change one selected feature and compare.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

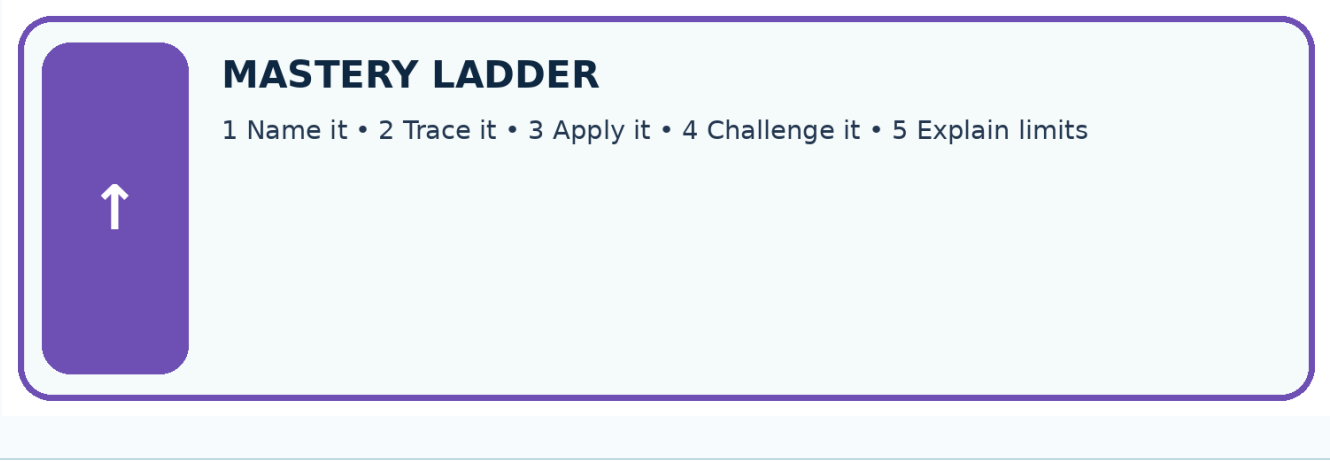
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Similarity by Selected Traits. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

Write or draw your reasoning and evidence.

STUDENT LESSON · 11 OF 18

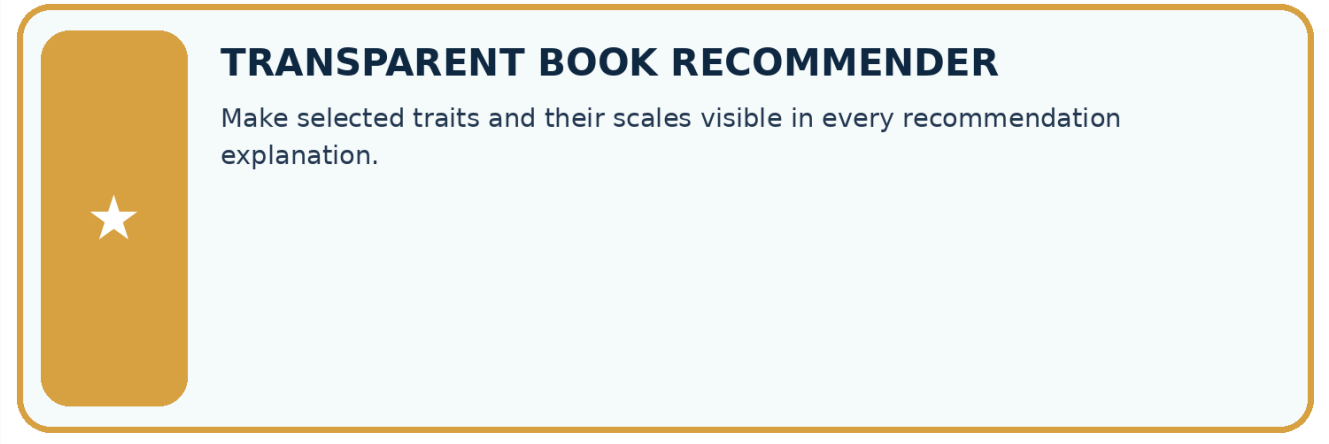
Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Make selected traits and their **scale**s visible in every recommendation explanation.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Specify recommender feature meanings, units, scaling and tie policy before explaining nearest matches.

STUDENT LESSON · 12 OF 18

Show what you know

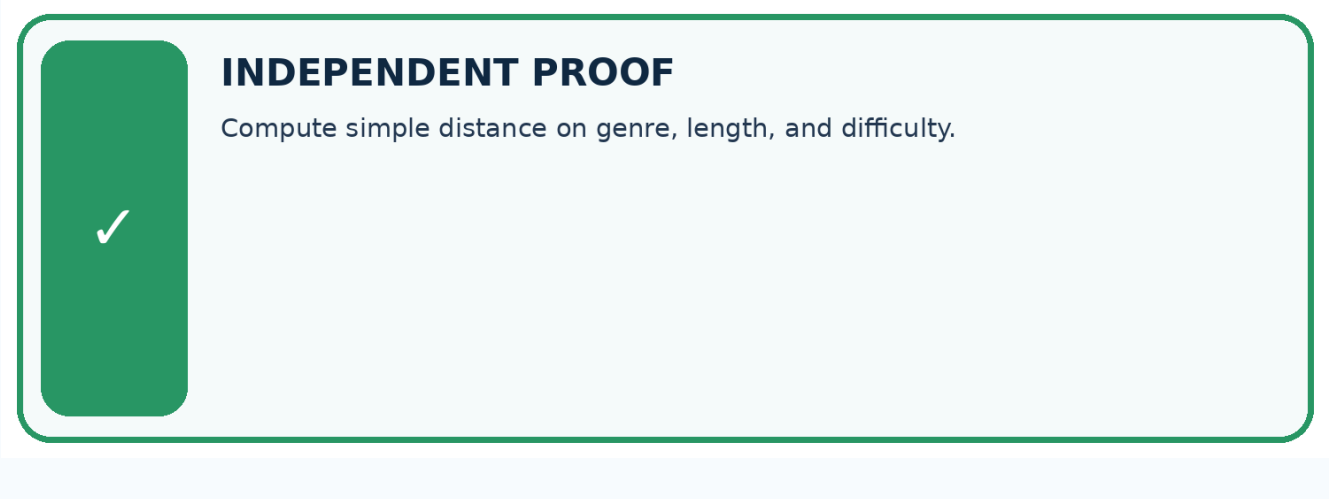
> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Compute simple **distance** on genre, length, and difficulty.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Similarity by Selected Traits case for supported practice. Repeated examples do not become fresh assessments.

Week 21 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Traits are [animals,adventure]. A=[1,1], B=[1,0], C=[0,0]; query=[1,0].

The question

Use number of differing traits as **distance**.

Worked explanation

Distances: A=1, B=0, C=1. B is nearest with equal feature weights.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Count differing traits from query [1,0]

Evidence or stage	Value or result
A [1,1]	1
B [1,0]	0
C [0,0]	1

Traits are [animals,adventure]. A=[1,1], B=[1,0], C=[0,0]; query=[1,0].

Use number of differing traits as **distance**.

Distances: A=1, B=0, C=1. B is nearest with equal feature weights.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Rank independently

Target is [3, 2]. Options A [1, 2], B [3, 4], C [4, 2]. Calculate **distance**s, select the closest, and identify a tie among the others.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 21 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Score **similarity** using one point for same topic and one for same difficulty. Target is space, easy. A is space, hard; B is animals, easy. What happens?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Make selected traits and their **scale**s visible in every recommendation explanation.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Represent books as [animal,short,illustrated] with 1=yes. Query=[1,1,0]; A=[1,0,0], B=[1,1,1], C=[0,0,0]. Count mismatches. What changes if illustrations matter twice as much?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Similarity by Selected Traits, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional toy book matcher represents two preferences with numeric features.

By the end, I can	How I will show it
Represent items with numeric features	Explain the deciding step, show a trace or test, and state one limit.
Calculate a simple distance	Explain the deciding step, show a trace or test, and state one limit.
Explain how units and chosen features affect similarity	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Similarity by Selected Traits without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Represent items with numeric features

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional toy book matcher represents two preferences with numeric features.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? B and C tie at **distance** 2; keep the tie visible.

1. **Distance** C

2. Choose closest

3. **Distance** A

4. **Distance** B

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Choose closest B and C tie at **distance** 2; keep the tie visible. Each **coordinate** contributes according to the declared **distance** rule.

Smaller **distance** means closer in this representation, not automatically a better reading experience. If one **coordinate** uses a much larger numerical **scale**, it can dominate the sum even when it is not more important.

Feature units and scaling belong in the method record.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Give the first-**coordinate** difference weight 2 while the second keeps weight 1. Which conclusion is supported by the supplied facts?

1. **Distance**s become A=3, B=4, C=3; A and C tie.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The

answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: **Distance**s become A=3, B=4, C=3; A and C tie. The representation's comparison rule changes; the supplied vectors stay the same.

Repair a missing building block

Review Repair Bias in Data (ai-2/20). Goal: Identify selection and coverage bias. Short practice: Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?.

Return to this week and explain what you changed.

Review Missing Data (ai-2/18). Goal: Detect absent and unusable values. Short practice: Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?. Return to this week and explain what you changed.

Review Choose Informative Questions (ai-2/15). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Choose the best original question and explain why its more balanced split matters here..

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

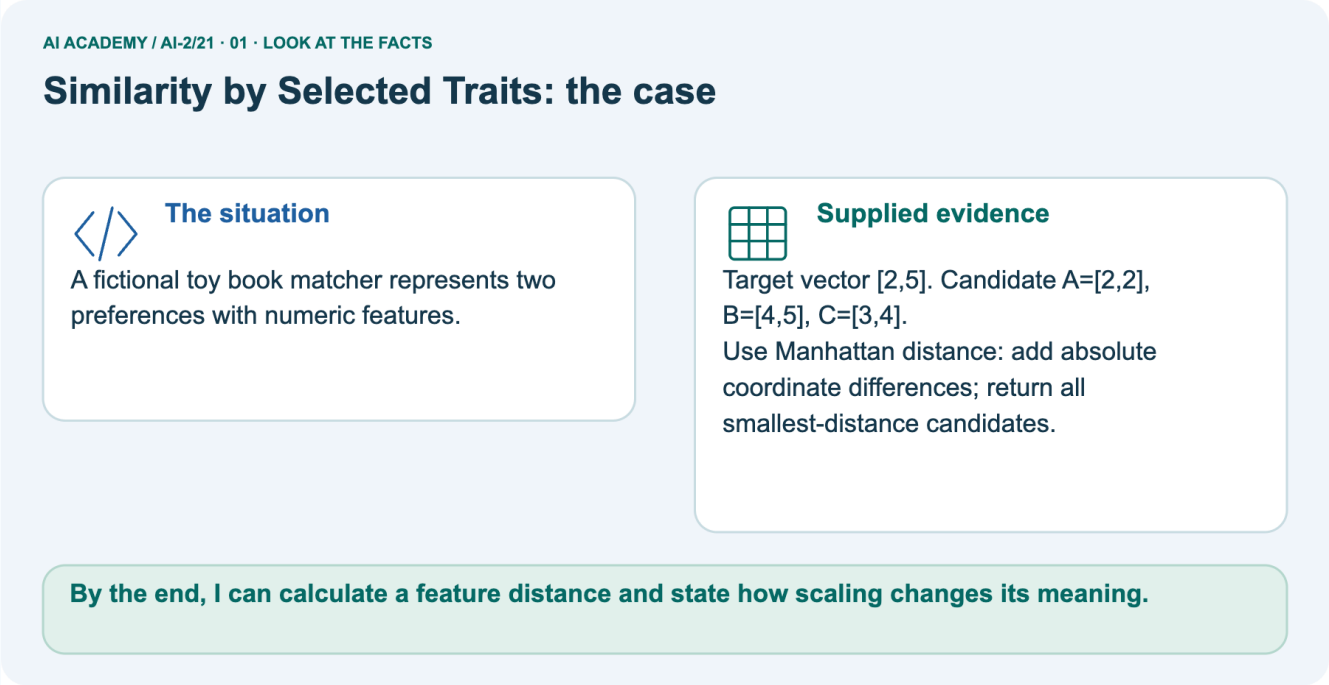
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Similarity by Selected Traits graphic using its labels, deciding evidence and relevant limitation.

01. Similarity by Selected Traits: the case



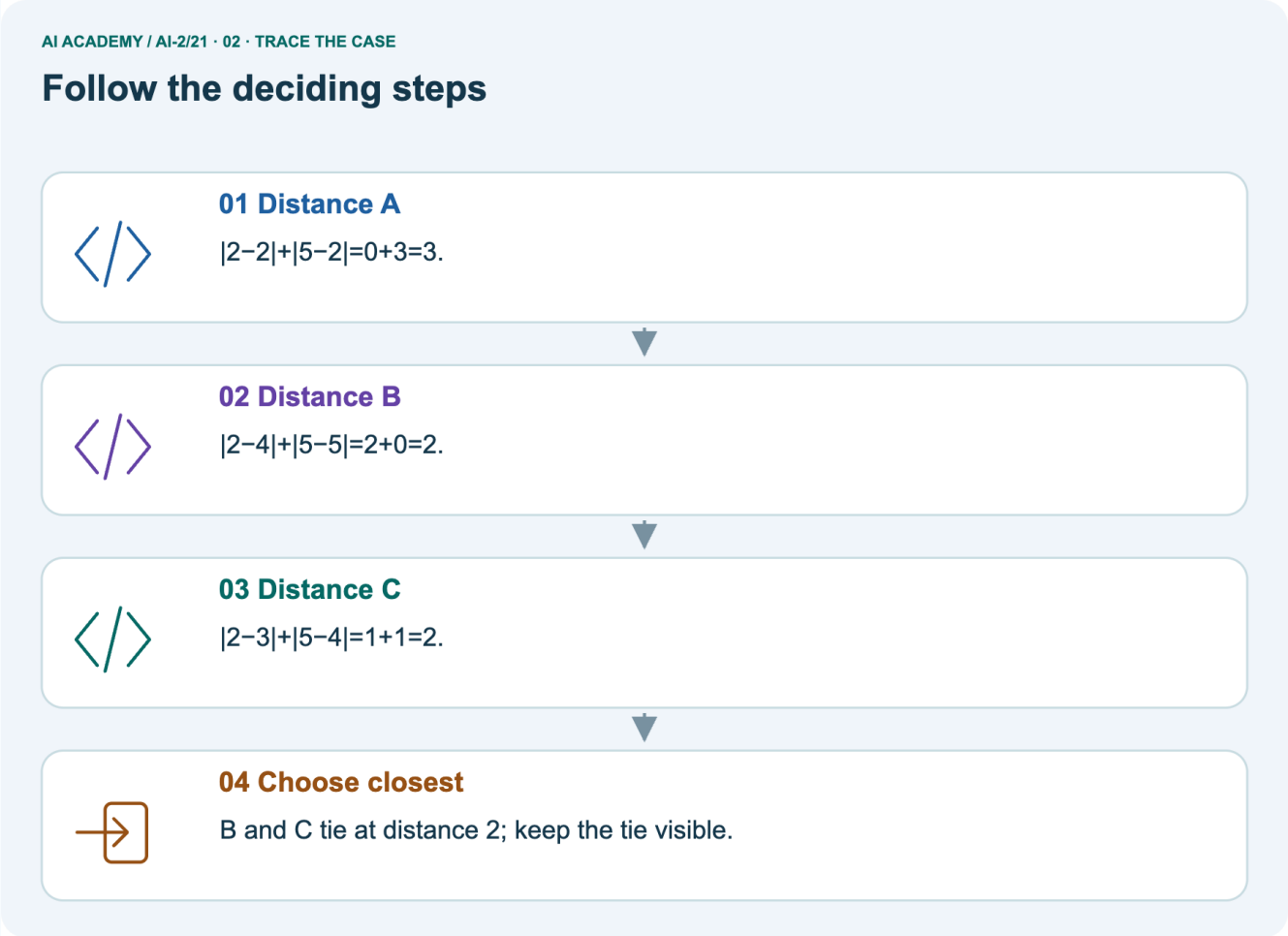
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional toy book matcher represents two preferences with numeric features. Supplied facts: Target vector [2,5]. Candidate A=[2,2], B=[4,5], C=[3,4].

Use **Manhattan distance**: add absolute **coordinate** differences; return all smallest-**distance** candidates.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

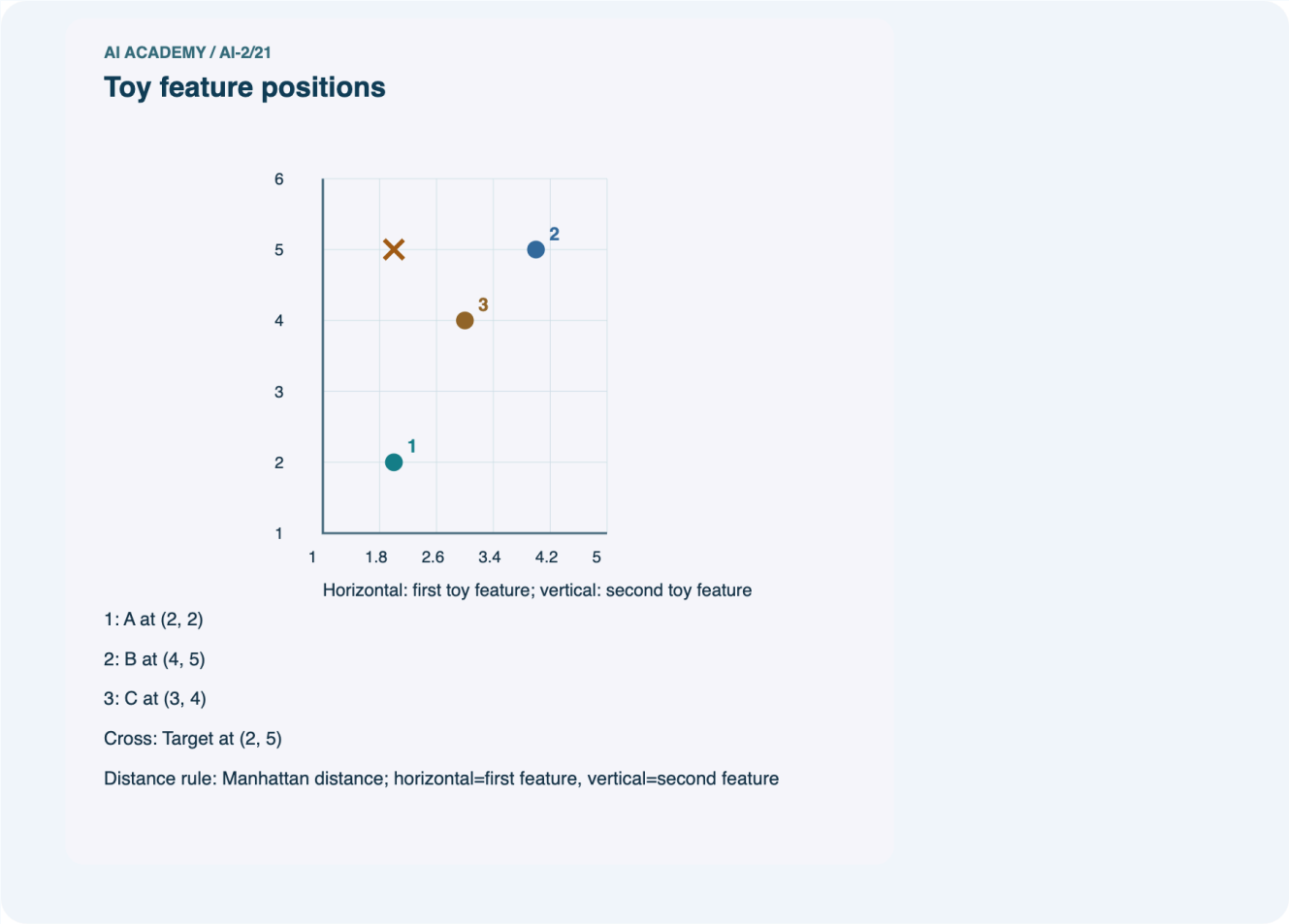


Follow the labelled picture with the matching explanation. Open the vector diagram

Distance A: |2−2|+|5−2|=0+3=3. → **Distance** B: |2−4|+|5−5|=2+0=2. → **Distance** C: |2−3|+|5−4|=1+1=2. → Choose closest: B and C tie at **distance** 2; keep the tie visible.

Point to each numbered stage and explain which recorded **input** it uses.

03. Toy feature positions

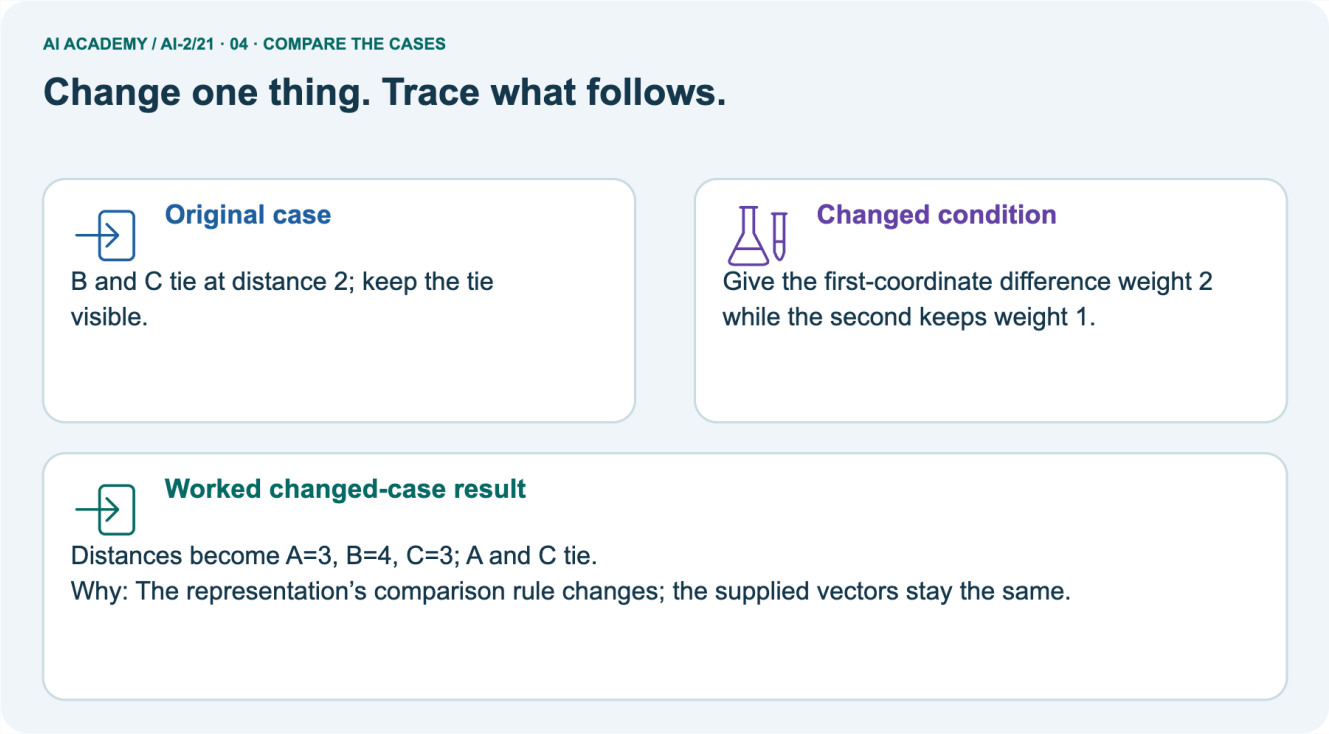


Follow the labelled picture with the matching explanation. Open the vector diagram

Toy feature positions. A: (2, 2). B: (4, 5). C: (3, 4). Query Target: (2, 5). Axes: first toy feature; second toy feature. **Distance**: **Manhattan distance**; horizontal=first feature, vertical=second feature.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



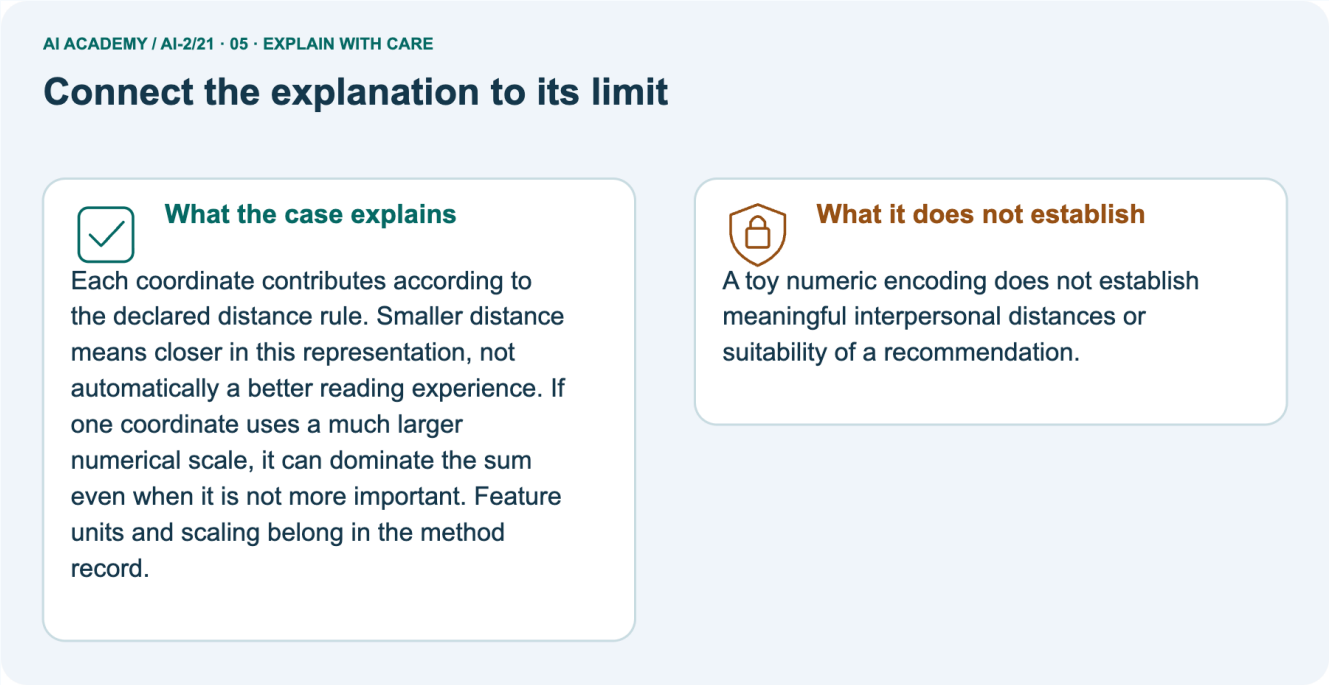
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: B and C tie at **distance** 2; keep the tie visible. Changed condition: Give the first-**coordinate** difference weight 2 while the second keeps weight 1. Expected result: **Distance**s become A=3, B=4, C=3; A and C tie.

Reason: The representation's comparison rule changes; the supplied vectors stay the same.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

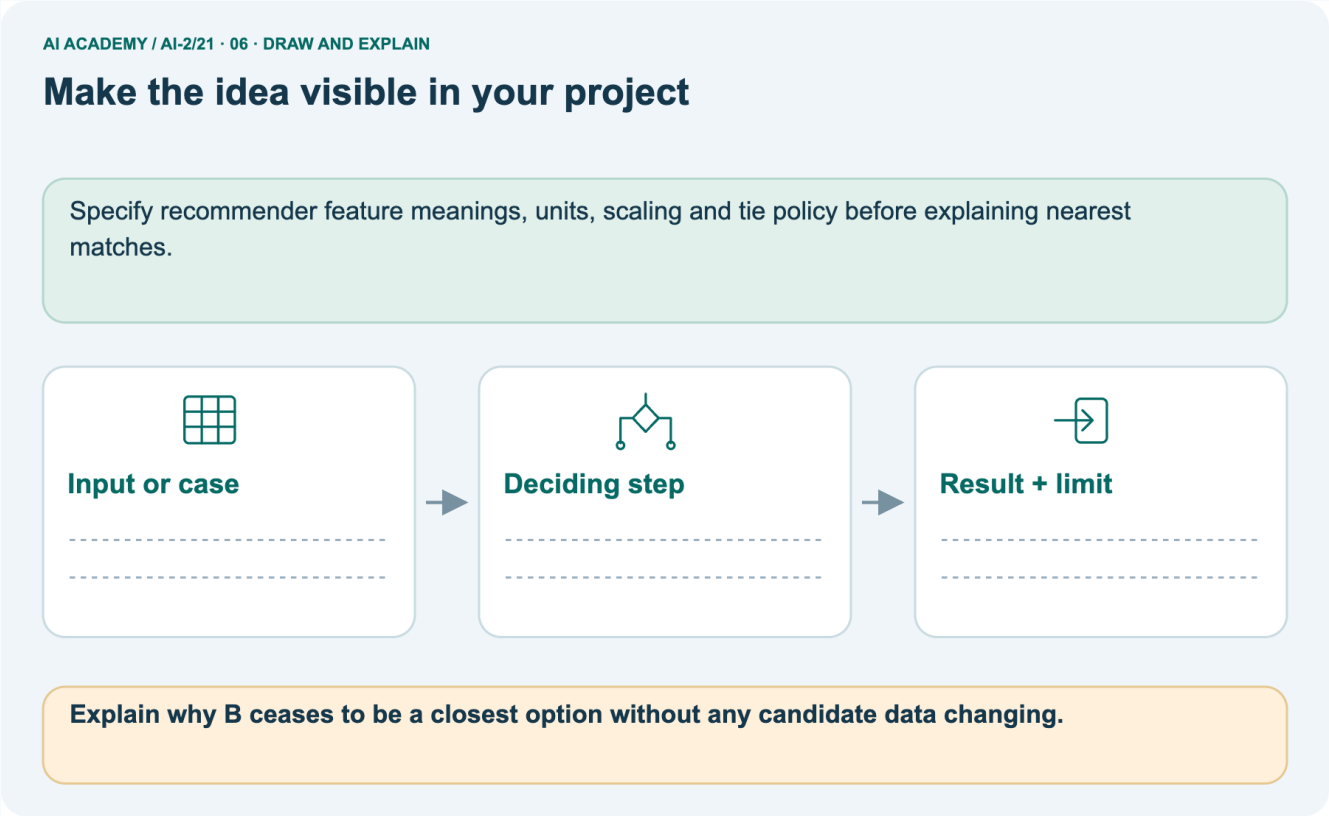
Each **coordinate** contributes according to the declared **distance** rule. Smaller **distance** means closer in this representation, not automatically a better reading experience.

If one **coordinate** uses a much larger numerical **scale**, it can dominate the sum even when it is not more important. Feature units and scaling belong in the method record.

Limit: A toy numeric encoding does not establish meaningful interpersonal **distance**s or suitability of a recommendation.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Specify recommender feature meanings, units, scaling and tie policy before explaining nearest matches. Draw your **input**, deciding step, result and limit.

Supported question: Explain why B ceases to be a closest option without any candidate data changing.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.